T10 — The Kummer Clock and the m-Power Gauge Hierarchy

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Abstract

Situation. T4 (Murillo Montero 2026, T4; T0 §5) introduced the **radical clock**: the multiquadratic field $K_N = \mathbb{Q}(\sqrt{p} : p \le N,\ p \text{ prime})$ generated by the unit-Helmert normalizers $\sqrt{k(k+1)}$ across the dimensional axis, with the structural theorem that K_N grows properly over K_{N-1} exactly when N is prime. T6 §8 sketched the m**-power closure** $\pi_m(x)_s = x_s^m / \sum_j x_j^m$ for integer $m \ge 3$, noting that the conductor selection rule (T3, T0 Theorem 3.5) generalizes from the cubic Amari–Chentsov vertex to the m-fold form $a_1 + \cdots + a_m \equiv c \pmod N$. T0 §1.3 catalogued the m**-power gauge corridor** $\{M_r : r \in \mathbb{R} \cup \{\pm\infty\}\}$ — the one-parameter family of power means indexed by gauge exponent — with named integer rungs from $m = -\infty$ (tropical inf) through $m = 0$ (geometric) to $m = +\infty$ (tropical sup). T7 §5 (Theorem 5.2) proved that the closure diagonal $\mathcal{M}_z(m, c)$ at integer m is a section of the ρ_m**-twisted line bundle** on $\mathbb{CP}^{N-1}$ under the Kummer cover deck group μ_N.

Gap. Three forward doors remain open: (a) the **Kummer clock** — the analog of T4's multiquadratic radical clock for general integer $m \ge 3$, generated by m-th roots over the cyclotomic field $\mathbb{Q}(\zeta_m)$ — whose growth criterion (analog of T4 Theorem 5.1's "prime N rungs") is more subtle for $m \ge 3$ than for $m = 2$; (b) the m**-power gauge hierarchy** sweep — a systematic catalog of named integer rungs $m \in \mathbb{Z}$ with their closure maps, polar pair structure, classical physical literatures, and Kummer-cover content; (c) the **higher-order physical content** at $m \in \{3, 4, -3, -4\}$: third-order optical nonlinearity ($\chi^{(3)}$), four-wave mixing, cubic Waring representations, dipole-dipole and van der Waals interactions, all of which sit at specific integer rungs of the gauge corridor.

Action. We sweep the m-power gauge hierarchy systematically across integer m, identifying the closure map, conductor selection rule, polar dual ($m \leftrightarrow -m$), Kummer cover deck group, and physical content at each rung. We define the **Kummer clock** $K_N^{(m)} := \mathbb{Q}(\zeta_m,\ \sqrt[m]{p} : p \le N,\ p \text{ prime})$ — the natural arithmetic ambient of the m-th power gauge — and analyze its growth structure, identifying both the cases where the analog of T4's prime-growth theorem extends ($m = 2$ classically, and a generalization to all m via Kummer theory) and the cases where new arithmetic enters non-trivially. We provide worked examples at $m = 3$ (cubic Kummer extensions over $\mathbb{Q}(\zeta_3)$), $m = 4$ (quartic over $\mathbb{Q}(i)$), and $m = -3$ (the polar partner). The earlier T5/T6 work at $m = \pm 1, \pm 2$ is positioned as the worked low-rung cases of this general hierarchy.

Result. Six core claims:

1. m**-power gauge corridor.** The corridor $\{M_r : r \in \mathbb{R} \cup \{\pm\infty\}\}$ of T0 §1.3 has named integer rungs at every $m \in \mathbb{Z}$, with closure map π_m, conductor selection rule (T0 Thm 3.5), and polar partner π_{-m} (T5; T0 Thm 2.2.4 generalized). The corridor is closed under reciprocal involution $r \leftrightarrow -r$ and limits to tropical algebra at $r \to \pm\infty$ (T0 §2.5).
2. m**-power closure and selection rule.** $\pi_m(x)_s = x_s^m / \sum_j x_j^m$ for $m \in \mathbb{Z} \setminus \{0\}$, with apex generator $\mathcal{M}_x(m, c) = G(x)^{-m} \widehat{x^m}_c$ obeying $a_1 + \cdots + a_m \equiv c \pmod N$ (T0 Theorem 3.5).
3. **Kummer cover at integer** m**.** The closure diagonal $\mathcal{M}_z(m, c)$ on complex amplitudes is a section of the ρ_m-twisted line bundle on $\mathbb{CP}^{N-1}$, with deck group μ_N acting by $\zeta \mapsto \zeta^{-m}$ (T7 Theorem 5.2). The cover is trivial iff $N \mid m$.
4. **Kummer clock definition.** $K_N^{(m)} := \mathbb{Q}(\zeta_m, \sqrt[m]{p} : p \le N,\ p \text{ prime})$ is the natural arithmetic ambient of the m-power gauge at dimensional level N, generalizing T4's multiquadratic radical field $K_N = K_N^{(2)}$. The base $\mathbb{Q}(\zeta_m)$ contributes the cyclotomic content of the m-th roots of unity; the Kummer tower $\sqrt[m]{p}$ adjoins the m-th roots of primes up to N.
5. **Kummer clock growth.** For $m = 2$, the radical clock $K_N^{(2)} = K_N$ grows properly over $K_{N-1}^{(2)}$ iff N is prime (T4 Theorem 5.1, restated as T0 Theorem 5.1). For $m \ge 3$, the growth criterion is more subtle: $K_N^{(m)}$ adjoins $\sqrt[m]{N}$ whenever N is prime, but the precise relationship to L^m-Helmert-like ladder structures requires forward analysis. We give a partial structure theorem and identify the open conjectural form.

6. **Worked examples at $m = 3, 4, -3$.** Explicit Kummer-clock fields, closure maps, and physical-content identifications at the cubic ($m = 3$, chirality / $\chi^{(3)}$ optics / four-wave mixing / Waring on cubes), quartic ($m = 4$, $\chi^{(4)}$ susceptibility / quartic potentials / ϕ^4 theory), and inverse-cubic ($m = -3$, dipole-dipole) rungs of the hierarchy.

Harvest. T4's multiquadratic radical clock is the $m = 2$ rung of a much larger **Kummer clock** structure that lives over the full m-power gauge corridor. Each integer-m rung carries its own Kummer-cyclotomic arithmetic, its own closure map and conductor selection rule, and its own polar dual via reciprocal involution. The cyclic-cyclotomic-radical compositum $\mathbb{Q}(\zeta_m,\ \sqrt[m]{p} : p \leq N)$ is the natural place where the apex's gauge arithmetic lives at each rung. Physical literatures (third-order nonlinear optics, Waring's problem on cubes, dipole-dipole forces) populate the corridor at specific integer rungs; the categorical-simplex framework provides the uniform organizing structure.

Keywords: m-power gauge, Kummer extension, cyclotomic field, multiquadratic field, radical clock, conductor selection rule, polar pair, L^m Helmert ladder, cubic gauge, quartic gauge, $\chi^{(3)}$ nonlinear optics, four-wave mixing, Waring's problem, dipole-dipole interaction, tropical limits, gauge corridor.

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The radical clock of T4 is the $m = 2$ rung. The full Kummer clock is the gauge corridor read through its arithmetic ambient.

Cross-references from prior papers

This paper builds on specific results from earlier papers of the ל-series; we list the dependencies explicitly.

Cited result	Used for
T0 §1.3 (gauge corridor catalog)	§1.1 corridor extension to all integer m
T0 §2.2 Theorem 2.2 (analytic properties of K_y, parity)	§1.2 Theorem 1.2 (polar pair across the corridor)
T0 §2.5 (tropical limits $m \to \pm\infty$)	§1.3 corridor compactification
T0 §3.3 Theorem 3.5 (closure diagonal selection rule)	§2.1 m-fold conductor selection rule

Cited result	Used for
T0 §3.3 Corollary 3.6 (polar/reciprocal case at negative integer m)	§2.2, §3 polar pair
T0 §5.2 Theorem 5.1 (multiquadratic radical clock at $m = 2$)	§4 Kummer clock definition (the $m = 2$ specialization)
T3 (cubic vertex selection rule at $n = 3, c = 0$)	§5.1 cubic gauge content
T4 (Helmert ladder and L^2 normalization)	§7.1 L^m-Helmert ladder forward door
T5 (polar pair M_r, M_{-r} at $\|r\| = 1$)	§3 polar pair generalized to all $\|m\|$
T6 §8 (m-power closure sketch)	§1.1 m-power closure for all integer m
T7 §5 Theorem 5.2 (Kummer cover at integer β)	§2.3 Kummer cover on complex amplitudes

T10 is the systematic sweep that ties together the m-dependent threads from T0/T3/T4/T5/T6/T7 into a single gauge-corridor structure with the Kummer clock as its arithmetic ambient.

Notation

We carry the notation of T0, T4, T5, T6, T7 and consolidate it for the m-power gauge hierarchy.

Symbol	Meaning
N	Number of categorical labels, ≥ 2.
m	Integer gauge exponent (or "rung"), $m \in \mathbb{Z}$.
x	Positive ledger $x : \mathbb{Z}/N \to \mathbb{R}_{>0}$.
$G(x)$	Geometric mean $\left(\prod_j x_j\right)^{1/N}$.
π_m	m-power closure: $\pi_m(x)_s = x_s^m / \sum_j x_j^m$ for $m \neq 0$.
M_r	Power mean at exponent r: $M_r(x) = \left(\frac{1}{N}\sum_j x_j^r\right)^{1/r}$ for $r \neq 0$.
$\mathcal{M}_x(\beta, c)$	T0 apex generator (real-positive sector).
$\mathcal{M}_z(\beta, c)$	T7 complex apex generator.
ζ_m	Primitive m-th root of unity $e^{2\pi i/m}$.
μ_m	Group of m-th roots of unity $\{\zeta_m^k : 0 \leq k < m\} \subset \mathbb{C}^\times$.
$\mathbb{Q}(\zeta_m)$	m-th cyclotomic field.
K_N	Multiquadratic field $\mathbb{Q}(\sqrt{p} : p \leq N,\ p \text{ prime})$ $(= K_N^{(2)})$.
$K_N^{(m)}$	Kummer clock field $\mathbb{Q}(\zeta_m,\ \sqrt[m]{p} : p \leq N,\ p \text{ prime})$ (Definition 4.1).
$\mathrm{mf}_m(n)$	m-th-power-free kernel of n: largest divisor b of n such that $n = a^m b$ for some $a \in \mathbb{Z}_{>0}$.
ρ_m	Kummer cover character $\rho_m(\zeta) = \zeta^{-m}$ on μ_N (T7 Theorem 5.2).

Convention. m ranges over all of $\mathbb{Z} \setminus \{0\}$ (with $m = 0$ being the geometric gauge limit, treated separately). Positive m gives the "additive substance is x^m" gauge; negative m gives the polar partner via reciprocal involution. Tropical limits at $m \to \pm\infty$ recover the max-plus / min-plus algebra (T0 §2.5).

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1. The m-power gauge corridor systematically

1.1 Definition and rungs

The **m-power gauge corridor** is the family $\{M_r : r \in \mathbb{R} \cup \{\pm\infty\}\}$ of power means on $\mathbb{R}^N_{>0}$ (T0 §1.3), with named integer rungs corresponding to specific classical structures. We catalog the integer rungs and their associated content.

Definition 1.1 (m-power closure). For each non-zero $m \in \mathbb{Z}$, the **m-power closure** is

$$\pi_m : \mathbb{R}^N_{>0} \setminus \{0\} \longrightarrow \Delta^{N-1}, \qquad x \longmapsto \Big(\frac{x_s^m}{\sum_j x_j^m}\Big)_{s=0}^{N-1}.$$

For $m = 0$ the closure degenerates; the natural geometric-gauge map is $x \mapsto x/G(x)$ followed by closure, equivalently $\log x \mapsto \log x - \overline{\log x}$ in the centered log-ledger framework of T0 §2.

Catalog of named integer rungs. Following T0 §1.3 and extending:

m	Rung name	Additive substance	Classical literature
$+\infty$	tropical sup	dominant coordinate	Maslov dequantization, tropical geometry, large deviations
4	quartic	x^4 — energy density squared	$\chi^{(4)}$ susceptibility (five-wave mixing), quartic potentials, ϕ^4 theory
3	cubic	x^3 — volume / chirality	Amari–Chentsov tensor at $n = 3$, Waring's problem, cubic forms
2	quadratic / RMS / Born	x^2 — energy / intensity	Fisher–Rao, Born rule, T6, L^2 Hilbert space
1	arithmetic / mass	x — mass	softmax, simplex closure, T2
0	geometric	$\log x$ — multiplicative scale	Aitchison clr, Amari e-flat chart
-1	harmonic	$1/x$ — reciprocal capacity	parallel resistance, T5, softmin
-2	inverse-quadratic	$1/x^2$ — compliance	inverse participation ratio
-3	inverse-cubic	$1/x^3$	dipole-dipole, van der Waals, inverse cube law
-4	inverse-quartic	$1/x^4$	ion-induced-dipole interaction (Stone 2013, Ch. 2)
$-\infty$	tropical inf	recessive coordinate	min-plus algebra

The integer rungs $m \in \{-\infty, \dots, -2, -1, 0, 1, 2, 3, 4, \dots, +\infty\}$ are doubly closed: by the reciprocal involution $r \leftrightarrow -r$ (T5; T0 §1.3 polar pair), and by the tropical limits $r \to \pm\infty$ (T0 §2.5).

1.2 Polar pair structure

T5 established the polar pair structure (M_r, M_{-r}) under the reciprocal involution $x_i \leftrightarrow x_i^{-1}$. Combined with the $r \leftrightarrow -r$ correspondence:

Theorem 1.2 (Polar pair across the corridor — T5 generalized). For every integer $m \neq 0$, the m-power gauge and the $-m$-power gauge are polar partners under the reciprocal involution $\iota : x \mapsto x^{-1}$:

$$\pi_m(x) = \pi_{-m}(\iota x) \quad \text{after coordinate identification } s \leftrightarrow s.$$

At the apex generator level (T0 §2.2 Theorem 2.2.4 generalized): the m-th and $-m$-th cumulant/closure samples of the apex $K_y(\beta)$ at $\beta = \pm m$ split into even and odd parity:

$$K_y(m) + K_y(-m) = 2\sum_{k\geq 1} \frac{\kappa_{2k}}{(2k)!} m^{2k} \quad \text{(even cumulants doubled)},$$

$$K_y(m) - K_y(-m) = 2\sum_{k\geq 1} \frac{\kappa_{2k+1}}{(2k+1)!} m^{2k+1} \quad \text{(odd cumulants doubled)}.$$

Proof. Direct from $\iota(x)_i = 1/x_i$, so $(\iota x)^m = x^{-m}$ componentwise; substituting into $\pi_m(\iota x)$ recovers $\pi_{-m}(x)$ via coordinate reciprocation. The parity statement is T0 Theorem 2.2.4 evaluated at $\beta = m$. □

The polar pair structure gives the corridor a $\mathbb{Z}/2$-symmetry around $m = 0$ (the geometric center), making each rung m structurally equivalent to its negative partner $-m$ up to reciprocation. T5 worked the $|m| = 1$ polar pair; T6 worked $|m| = 2$; T10 makes the structure explicit at all integer $|m| \geq 3$.

1.3 Tropical limits at $m \to \pm\infty$

T0 §2.5 (Theorem 2.2.5) gave the tropical limit:

$$\lim_{m\to+\infty} \frac{1}{m} K_y(m) = \max_i y_i, \qquad \lim_{m\to-\infty} \frac{1}{m} K_y(m) = \min_i y_i.$$

In the gauge corridor, large $|m|$ approaches the tropical extremes — winner-take-all dynamics for $m \to +\infty$, loser-take-all for $m \to -\infty$. These limits are the categorical analog of phase transitions to "tropical statistics" (max-plus / min-plus algebra in the sense of Maslov 1987, Litvinov–Maslov 2005).

The integer rungs $m \in \mathbb{Z}$ are samples of this continuous corridor; the Kummer-clock arithmetic developed in §4 lives at each integer rung individually.

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2. The m-power closure and conductor selection rule

2.1 Closure and apex generator

By T0 Theorem 3.5 (specialized to real-positive ledgers; T7 §3 lifts to complex), the apex generator of the m-power closure at character index $c \in \widehat{\mathbb{Z}/N}$ is

$$\mathcal{M}_x(m, c) = G(x)^{-m} \widehat{x^m}_c = G(x)^{-m} \cdot \frac{1}{N} \sum_{s\in\mathbb{Z}/N} x_s^m \, \overline{\chi_c(s)},$$

with the convolution-form expansion

$$\widehat{x^m}_c = \sum_{\substack{a_1,\ldots,a_m \in \widehat{\mathbb{Z}/N} \\ a_1+\cdots+a_m \equiv c \,(\mathrm{mod}\, N)}} \hat{x}_{a_1} \hat{x}_{a_2} \cdots \hat{x}_{a_m},$$

the m**-fold conductor selection rule**.

Examples by m: - $m = 1$ (linear): $a_1 \equiv c \pmod N$, single-index identity (trivial selection). - $m = 2$ (Born): $a_1 + a_2 \equiv c \pmod N$ — T6 RMS-conductor mixing rule. - $m = 3$ (cubic): $a_1 + a_2 + a_3 \equiv c \pmod N$ — T3 cubic vertex selection rule (recovers Amari–Chentsov at $c = 0$). - $m = 4$ (quartic): $a_1 + a_2 + a_3 + a_4 \equiv c \pmod N$ — quartic conductor mixing ($\chi^{(4)}$ susceptibility, five-wave mixing process).

2.2 Reciprocal pair via Corollary 3.6

T0 Corollary 3.6 covered the negative-integer case via the reciprocal ledger $u := 1/x$:

$$\mathcal{M}_x(-m, c) = \mathcal{M}_u(m, c) \quad \text{with selection rule on } \hat{u}\text{-coefficients.}$$

So the negative-m rungs are accessible via reciprocation, with their own selection rules involving $\widehat{x^{-m}}_c$ — equivalently, $\widehat{u^m}_c$ where $u = 1/x$.

2.3 Integer m via T7's Kummer cover

For complex amplitudes, T7 Theorem 5.2 gave the full Galois statement: $\mathcal{M}_z(m, c)$ at integer m is a section of the ρ_m-twisted line bundle on $\mathbb{CP}^{N-1}$, with the Kummer cover deck group μ_N acting by $\zeta \mapsto \zeta^{-m}$. The cover is trivial iff $N \mid m$.

For the real-positive sector (T7 Proposition 3.4 and T0 Proposition 3.7), the Kummer cover trivializes, recovering T0's apex generator.

This gives the **integer-m Galois structure** of each gauge rung uniformly across m.

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3. The polar-pair $m \leftrightarrow -m$ in detail

For each positive integer m, the $-m$ rung is the polar partner. The polar pair structure was developed in T5 for $|m| = 1$ and in T6/T0 §2.4 for $|m| = 2$; it extends uniformly to all $|m|$ via the reciprocal involution.

Structural summary of the $(m, -m)$ pair for general integer $m \geq 1$:

Feature	m rung	$-m$ rung	Pair structure
Closure	$\pi_m(x)_s = x_s^m / \sum x^m$	$\pi_{-m}(x)_s = x_s^{-m} / \sum x^{-m}$	Reciprocal involution
Additive substance	x^m	$1/x^m$	Reciprocal
Apex sample	$\mathcal{M}_x(m, c)$	$\mathcal{M}_x(-m, c) = \mathcal{M}_{1/x}(m, c)$	Reciprocal-ledger substitution
Selection rule	$a_1 + \cdots + a_m \equiv c \pmod N$	$a_1 + \cdots + a_m \equiv c \pmod N$ on $\widehat{x^{-1}}$	Same indices, different coefficients
Cumulant content	$K_y(m)$	$K_y(-m)$	Even-cumulant sum, odd-cumulant difference
Physical example ($m = 3$)	$\chi^{(3)}$ optics, four-wave mixing	dipole-dipole, van der Waals	Forward vs reverse interaction

The $\mathbb{Z}/2$-symmetry of the gauge corridor around $m = 0$ identifies the structure of the m and $-m$ rungs as **dual interpretations** of the same underlying cumulant pair $(\kappa_{2k}^{\text{even}}, \kappa_{2k+1}^{\text{odd}})$, with the "direct" interpretation at positive m and the "reciprocal" interpretation at negative m.

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4. The Kummer clock

4.1 Definition and base structure

Definition 4.1 (Kummer clock). For integer $m \geq 2$ and dimensional level $N \geq 2$, the **Kummer clock field** at level (m, N) is

$$K_N^{(m)} := \mathbb{Q}(\zeta_m,\ \sqrt[m]{p} : p \leq N,\ p \text{ prime}),$$

the compositum of the m-th cyclotomic field $\mathbb{Q}(\zeta_m)$ with the Kummer tower $\mathbb{Q}(\sqrt[m]{p} : p \le N)$ over primes up to N.

For $m = 2$: $\zeta_2 = -1 \in \mathbb{Q}$, so $\mathbb{Q}(\zeta_2) = \mathbb{Q}$, and $K_N^{(2)} = \mathbb{Q}(\sqrt{p} : p \le N) = K_N$ recovers T4's multiquadratic radical clock.

For $m \ge 3$: $\mathbb{Q}(\zeta_m)$ is a proper extension of $\mathbb{Q}$, of degree $\varphi(m)$, and the Kummer tower over $\mathbb{Q}(\zeta_m)$ is a (potentially) higher-degree structure.

4.2 Degree bounds via Kummer theory

By standard Kummer theory (Lang, *Algebra* VI §6): over a base field K containing μ_m, the extension $K(\sqrt[m]{a})$ for $a \in K^\times$ is cyclic of degree dividing m, with Galois group a subgroup of μ_m. For independent generators $a_1, \dots, a_r \in K^\times/(K^\times)^m$, the compositum $K(\sqrt[m]{a_1}, \dots, \sqrt[m]{a_r})$ has degree m^r over K provided the a_i are multiplicatively independent modulo m-th powers.

Proposition 4.2 (Kummer clock degree bound). For prime m and $N \ge 2$:

$$[K_N^{(m)} : \mathbb{Q}(\zeta_m)] \le m^{\pi(N)}, \qquad [K_N^{(m)} : \mathbb{Q}] \le \varphi(m) \cdot m^{\pi(N)},$$

where $\pi(N)$ is the prime-counting function. Equality holds when the rational primes $\{p : p \le N\}$ are multiplicatively independent modulo m-th powers in $\mathbb{Q}(\zeta_m)^\times$, which is the generic case (and a classical theorem for $m = 2$ — Kronecker's multiquadratic linear independence).

Proof. Standard Kummer theory plus the structure-theorem upper bound on Galois group orders. Multiplicative independence of rational primes mod m-th powers in $\mathbb{Q}(\zeta_m)^\times$ requires verifying that no non-trivial product $\prod p_i^{e_i}$ ($0 < e_i < m$, primes $\le N$) is an m-th power in $\mathbb{Q}(\zeta_m)$. For $m = 2$, this is the classical Kronecker result used in T4 Theorem 5.1 / T0 Theorem 5.1. For $m \ge 3$, the analogous result is generic but requires checking specific arithmetic relations on a case-by-case basis. □

For $m = 2$: the bound is tight (equality), giving $[K_N : \mathbb{Q}] = 2^{\pi(N)}$ as in T4.

For $m = 3$: the bound is tight for generic N but may fail at special N where unexpected cubic relations hold; the analog of Kronecker linear independence for cube roots over $\mathbb{Q}(\zeta_3)$ is a classical fact of Kummer theory (Lang, *Algebra* VI §6; Washington, *Introduction to Cyclotomic Fields*, Ch. 10 — that $\sqrt[3]{2}, \sqrt[3]{3}, \sqrt[3]{5}, \sqrt[3]{7}, \dots$ are linearly independent over $\mathbb{Q}(\zeta_3)$, the cube-root analog of the Besicovitch (1940) result for square roots over $\mathbb{Q}$). This gives the full degree $3^{\pi(N)}$ for $K_N^{(3)}$ over $\mathbb{Q}(\zeta_3)$, hence $[K_N^{(3)} : \mathbb{Q}] = \varphi(3) \cdot 3^{\pi(N)} = 2 \cdot 3^{\pi(N)}$.

4.3 Growth criterion (partial)

T4 Theorem 5.1 (T0 Theorem 5.1) gave the **prime-rung growth criterion** for $m = 2$: $K_N^{(2)} \supsetneq K_{N-1}^{(2)}$ iff N is prime.

For $m \ge 3$, an analogous criterion is more subtle.

Proposition 4.3 (Necessary condition for Kummer clock growth). $K_N^{(m)} \supsetneq K_{N-1}^{(m)}$ requires either (a) N is prime (so $\sqrt[m]{N}$ is a new generator), or (b) some non-trivial Kummer-theoretic identity involving primes $\le N$ relative to the structure of $K_{N-1}^{(m)}$ produces a new element.

Proof. The new generator added at rung N over $\mathbb{Q}(\zeta_m)$ is $\sqrt[m]{N}$. If N is composite with prime factorization $N = \prod p_i^{a_i}$, then $\sqrt[m]{N} = \prod p_i^{a_i/m}$ is a product of (rational) integer powers and m-th roots of primes that all lie in $K_{N-1}^{(m)}$ (since the primes p_i are all $\le N-1$ for composite N with smallest prime factor $p \le \sqrt{N} \le N-1$, except for $N = 4$ where $\sqrt[m]{4} = \sqrt[m]{2^2} = 2^{2/m}$ which is $\sqrt{2}$ for $m = 4$, $\sqrt[3]{4}$ for $m = 3$, etc. — handled by case analysis). If N is prime, then $\sqrt[m]{N}$ is a new m-th root not in $K_{N-1}^{(m)}$, so the extension is proper. □

So the **necessary** condition is "N is prime"; the **sufficient** condition follows when the m-th roots of primes are multiplicatively independent (Proposition 4.2). For generic m, this gives a parallel to T4's prime-growth criterion. The precise growth structure for $m \geq 3$ at small N requires case-by-case analysis.

4.4 The Kummer clock as a $\mathbb{Z} \times \mathbb{Z}_{\geq 2}$-graded tower

Across both the dimensional axis (N) and the gauge-exponent axis (m), the Kummer clock forms a **double tower**:

$$\{K_N^{(m)} : m \in \mathbb{Z}_{\geq 2},\ N \in \mathbb{Z}_{\geq 2}\}$$

with inclusions $K_{N-1}^{(m)} \subseteq K_N^{(m)}$ (dimensional growth at fixed m) and $K_N^{(m_1)} \to K_N^{(m_2)}$ for $m_1 \mid m_2$ (cyclotomic-Kummer compatibility along the gauge axis when m_1 divides m_2).

The double tower exhibits: - **Horizontal growth** along N: governed by prime-rung criterion (Proposition 4.3). - **Vertical growth** along m: governed by the cyclotomic content $\mathbb{Q}(\zeta_m)$ and the Kummer-tower content $\sqrt[m]{p}$. - **Diagonal interactions**: e.g., $K_4^{(2)} = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ vs $K_4^{(3)} = \mathbb{Q}(\zeta_3, \sqrt[3]{2}, \sqrt[3]{3})$ — different arithmetic ambient at different gauge rungs.

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5. Worked examples by gauge rung m

5.1 $m = 3$: the cubic gauge

Kummer clock at $m = 3$. $K_N^{(3)} = \mathbb{Q}(\zeta_3, \sqrt[3]{p} : p \leq N)$ where $\zeta_3 = e^{2\pi i/3} = (-1 + i\sqrt{3})/2$.

For small N: - $N = 2$: $K_2^{(3)} = \mathbb{Q}(\zeta_3, \sqrt[3]{2})$, degree $2 \cdot 3 = 6$ over $\mathbb{Q}$ (since 2 is the only prime ≤ 2). - $N = 3$: $K_3^{(3)} = \mathbb{Q}(\zeta_3, \sqrt[3]{2}, \sqrt[3]{3})$, degree $2 \cdot 9 = 18$ (multiplicatively independent cube roots). - $N = 5$: $K_5^{(3)} = \mathbb{Q}(\zeta_3, \sqrt[3]{2}, \sqrt[3]{3}, \sqrt[3]{5})$, degree $2 \cdot 27 = 54$. - $N = 4$ (composite): $\sqrt[3]{4} = \sqrt[3]{2^2} = (\sqrt[3]{2})^2 \in K_3^{(3)}$, so $K_4^{(3)} = K_3^{(3)}$, no growth.

The growth criterion (Proposition 4.3) is exact at $m = 3$ for these small N.

Apex content at $m = 3$. The closure diagonal $\mathcal{M}_x(3, c) = G(x)^{-3}\widehat{x^3}_c$ obeys the cubic selection rule $a_1 + a_2 + a_3 \equiv c \pmod{N}$ (T3 cubic vertex generalized from the Amari–Chentsov tensor).

Kummer cover at $m = 3$. By T7 Theorem 5.2, on complex amplitudes, $\mathcal{M}_z(3, c)$ transforms under the deck group μ_N by $\zeta \mapsto \zeta^{-3}$. The cover is trivial iff $N \mid 3$, i.e., for $N \in \{1, 3\}$; for all other N the cover is non-trivial, with deck-group character of order $N/\gcd(N, 3)$.

Physical content. The cubic rung carries: - $\chi^{(3)}$ **nonlinear optics**: third-order susceptibility tensors in the optical response of materials produce four-wave mixing terms cubic in the field amplitude. The conductor selection rule $a + b + c \equiv 0 \pmod{N}$ matches the energy-momentum conservation in $3 \to 1$ photon mixing. - **Waring's problem on cubes**: every positive integer is the sum of at most 9 non-negative cubes ($g(3) = 9$; Wieferich 1909, completing Hilbert), and every sufficiently large integer is the sum of at most $G(3) \leq 7$ cubes (Linnik 1943); the conjectural value $G(3) = 4$ remains open. The categorical-simplex apex at $m = 3$ encodes the cubic-form arithmetic at the level of conductor packets. - **Cubic chirality**: the Amari–Chentsov skewness tensor (T2, T3) is the apex's first non-trivial cubic invariant, with the conductor selection rule fixing chirality content modulo N.

5.2 $m = 4$: the quartic gauge

Kummer clock at $m = 4$. $K_N^{(4)} = \mathbb{Q}(\zeta_4, \sqrt[4]{p} : p \leq N) = \mathbb{Q}(i, \sqrt[4]{p} : p \leq N)$.

For small N: - $N = 2$: $K_2^{(4)} = \mathbb{Q}(i, \sqrt[4]{2})$, degree $2 \cdot 4 = 8$. - $N = 3$: $K_3^{(4)} = \mathbb{Q}(i, \sqrt[4]{2}, \sqrt[4]{3})$, degree $2 \cdot 16 = 32$. - $N = 4$: $\sqrt[4]{4} = \sqrt{2} = (\sqrt[4]{2})^2 \in K_3^{(4)}$, so $K_4^{(4)} = K_3^{(4)}$, no growth. - $N = 5$: $K_5^{(4)} = \mathbb{Q}(i, \sqrt[4]{2}, \sqrt[4]{3}, \sqrt[4]{5})$, degree $2 \cdot 64 = 128$.

Apex content at $m = 4$. Selection rule $a_1 + a_2 + a_3 + a_4 \equiv c \pmod N$ — quartic conductor mixing.

Kummer cover at $m = 4$. Trivial iff $N \mid 4$, i.e., $N \in \{1, 2, 4\}$.

Physical content. The quartic rung carries: - $\chi^{(4)}$ **susceptibility / five-wave mixing**: the next-order nonlinear-optics process after $\chi^{(3)}$ (which sits at $m = 3$, §5.1), with quartic substance in the field amplitude and a 5-photon energy-momentum diagram (4 inputs + 1 output) matching the conductor selection rule $a_1 + a_2 + a_3 + a_4 \equiv c \pmod N$ at $m = 4$. - **Quartic confining potentials**: $V(x) \sim x^4$ — anharmonic oscillators, ϕ^4 field theory at the Lagrangian level. - The polar partner $m = -4$ corresponds to $\sim 1/r^4$ potentials, characteristic of the **ion-induced-dipole** interaction in multipole expansions (Stone 2013, Ch. 2). The deeper-tail polynomial decays $1/r^6$ (London dispersion) and $1/r^{12}$ (the Lennard-Jones 12-6 phenomenological repulsion; the Born–Mayer repulsion $V(r) = A\exp(-r/\rho)$ is *exponential*, not polynomial, and does not sit on the gauge corridor at any integer m) sit at deeper negative rungs ($m = -6$, $m = -12$), not at $m = -4$.

5.3 $m = -3$: the inverse-cubic polar partner

Kummer clock at $m = -3$. By Theorem 1.2 polar pair structure, the -3 rung is the reciprocal partner of $+3$. The Kummer clock structure: $K_N^{(-3)} = K_N^{(3)}$ (the field of cube roots is the same, just the gauge is read reciprocally).

Apex content at $m = -3$. $\mathcal{M}_x(-3, c) = \mathcal{M}_{1/x}(3, c)$ — the cubic apex of the reciprocal ledger. Selection rule: $a_1 + a_2 + a_3 \equiv c \pmod N$ on $\widehat{(1/x)}$ — same indices as $m = +3$, but on the reciprocal Fourier coefficients.

Physical content. The inverse-cubic rung carries: - **Dipole-dipole interaction**: the long-range force between two electric dipoles falls as $1/r^3$ — the $m = -3$ rung of the gauge corridor. - **van der Waals (Keesom) forces**: orientational dipole-dipole averaging gives $1/r^6$, polar partner at $m = -6$. - **Coulomb potential**: $1/r$ is the $m = -1$ rung (harmonic / reciprocal); the -3 rung is the next-order multipole contribution.

The polar dual pair $(+3, -3)$ exhibits **cubic chirality** (forward, $m = +3$) and **cubic compliance** (reverse, $m = -3$) as dual interpretations of the same conductor selection rule on the apex.

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6. The gauge corridor as a unifying organizing principle

6.1 What lives at each rung

Bringing the catalog together, each integer rung m of the gauge corridor carries:

1. **Closure map** π_m (Definition 1.1).
2. **Conductor selection rule** $a_1 + \cdots + a_m \equiv c \pmod N$ on the apex generator (T0 Thm 3.5).
3. **Polar partner** $-m$ via reciprocal involution (Theorem 1.2).
4. **Kummer cover** on complex amplitudes with deck-group character $\rho_m(\zeta) = \zeta^{-m}$ (T7 Thm 5.2).
5. **Kummer clock field** $K_N^{(m)}$ encoding the arithmetic ambient (Definition 4.1).
6. **Classical literature(s)** — physics, statistics, number theory — populated at the rung (§5).

6.2 Worked rung summary table

m	Closure	Selection rule	Polar partner	Kummer cover trivial iff	Kummer clock	Physical content
m	Closure	Selection rule	Polar partner	Kummer cover trivial iff	Kummer clock	Physical content
—	—	—	—	—	—	—
1	mass / simplex	$a_1 \equiv c \pmod N$ (no convolution)	-1 (harmonic)	$N \mid 1$, i.e., $N = 1$ (never for $N \geq 2$)	$K_N^{(1)} = \mathbb{Q}$	T2 cumulant tower
2	Born / RMS	$a_1 + a_2 \equiv c \pmod N$	-2 (compliance)	$N \mid 2$, i.e., $N \in \{1, 2\}$	$K_N^{(2)} = K_N$	T6, Fisher–Rao, quantum probability
3	cubic	$a_1+a_2+a_3 \equiv c \pmod N$	-3 (dipole-dipole)	$N \mid 3$, i.e., $N \in \{1, 3\}$	$K_N^{(3)}$	T3 cubic vertex, $\chi^{(3)}$ optics, Waring
4	quartic	$a_1 + a_2 + a_3 + a_4 \equiv c \pmod N$	-4 (ion-induced-dipole)	$N \mid 4$, i.e., $N \in \{1, 2, 4\}$	$K_N^{(4)}$	$\chi^{(4)}$ (five-wave mixing), ϕ^4 theory
$m \geq 5$	higher-order	$\sum_{i=1}^m a_i \equiv c \pmod N$	$-m$	$N \mid m$	$K_N^{(m)}$	higher harmonic generation
-1	harmonic	$a_1 \equiv c \pmod N$ (on $\widehat{x^{-1}}$)	1	$N \mid 1$, i.e., $N = 1$ (never for $N \geq 2$)	$K_N^{(-1)} = \mathbb{Q}$	T5, parallel resistance
$m \leq -2$	reciprocal $\lvert m\rvert$	$\sum_{i=1}^{\lvert m\rvert} a_i \equiv c$ on $\widehat{x^{-\lvert m\rvert}}$	$\lvert m\rvert$	$N \mid \lvert m\rvert$	$K_N^{(\lvert m\rvert)}$	multipole interactions
$\pm\infty$	tropical	$\arg\max$ / $\arg\min$	$\mp\infty$	n/a (tropical limit, no Kummer structure)	$\mathbb{Q}$	LDT, max-plus algebra

The corridor is a uniform structure indexed by $m \in \mathbb{Z} \cup \{\pm\infty\}$, with the Kummer clock field $K_N^{(m)}$ providing the natural arithmetic ambient at each rung.

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7. Forward doors

7.1 L^m-Helmert ladder

T4's L^2-Helmert ladder uses the contrasts $h_k = (1, \ldots, 1, -k, 0, \ldots, 0)$ normalized by $\sqrt{k(k+1)} = \|h_k\|_2$. The

analog for general m: normalize the same contrasts by $\|h_k\|_m = (k + k^m)^{1/m} = k^{1/m}(1 + k^{m-1})^{1/m}$, giving unit-$L^m$ Helmert vectors $h_k/\|h_k\|_m$.

The Kummer-clock content of these L^m-normalizers: $\sqrt[m]{k(1+k^{m-1})}$ for $m \geq 2$. The analog of T4 Theorem 5.1's prime-growth theorem in this L^m-Helmert setting is conjectural: does $\mathbb{Q}(\sqrt[m]{\mathrm{mf}_m(k(1+k^{m-1}))} : 1 \leq k < N) = K_N^{(m)}$?

A direct check at $m = 3, N \leq 4$: $\|h_1\|_3^3 = 2$, $\|h_2\|_3^3 = 10$, $\|h_3\|_3^3 = 30$. Field generated: $\mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{10}, \sqrt[3]{30}) = \mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{5}, \sqrt[3]{3})$ (since $\sqrt[3]{10} = \sqrt[3]{2}\sqrt[3]{5}$ and $\sqrt[3]{30} = \sqrt[3]{2}\sqrt[3]{3}\sqrt[3]{5}$). This equals $K_3^{(3)}$ up to the cyclotomic content ζ_3 which L^m-normalizers themselves do not provide. So the L^m-Helmert ladder generates the **real** Kummer subfield; the cyclotomic content must be added separately.

The precise growth criterion and full structure theorem for the L^m-Helmert ladder is a forward door.

Forward paper title (planned): **T10.1 — The L^m-Helmert Ladder.**

7.2 Non-integer gauge $\beta \in \mathbb{R} \setminus \mathbb{Z}$

The gauge corridor is continuous in $\beta \in \mathbb{R}$, but T10 has focused on integer rungs $m \in \mathbb{Z}$ where the closure map is polynomial and the Kummer cover is finite. For non-integer β, the apex generator $\mathcal{M}_x(\beta, c)$ remains well-defined (T0 §2.2 Theorem 2.2) but the Kummer-cover structure becomes infinite-dimensional (universal cover via complex logarithm, T7 §3.4).

Forward paper title (planned): **T10.2 — The Non-Integer Gauge Continuum.**

7.3 Fractional gauge $m = 1/n$

A particularly interesting non-integer subcase: fractional rungs $\beta = 1/n$ for integer n, corresponding to the **n-th root** closure $x \mapsto x^{1/n}/\sum x^{1/n}$. These are connected to AJLS (Ay–Jost–Lê–Schwachhöfer 2017) half-densities ($n = 2$) and cube-root densities ($n = 3$), and play the role of "Kummer cover roots downstairs" in the categorical-simplex picture.

Forward paper title (planned): **T10.3 — Fractional Gauges and AJLS Density Powers.**

7.4 Physical applications at higher rungs

A systematic catalog of physical applications at integer rungs $|m| \geq 3$ — $\chi^{(3)}$ optics, $\chi^{(5)}$ optics, higher-order multipole interactions, anharmonic potentials in molecular physics, virial-equation-of-state coefficients — would substantiate the "physical literatures populating the corridor" theme of §5/§6.

Forward paper title (planned): **T10.4 — Physical Applications of the Gauge Corridor.**

7.5 Galois-group structure of the Kummer clock

The Galois group $\mathrm{Gal}(K_N^{(m)}/\mathbb{Q})$ has a structure that generalizes T4's $(\mathbb{Z}/2)^{\pi(N)}$ for $m = 2$. For general m, the structure is a semi-direct product involving $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) = (\mathbb{Z}/m)^\times$ and the Kummer-tower piece $(\mathbb{Z}/m)^{\pi(N)}$. The precise group-theoretic structure and its action on the apex generator is a forward door.

Forward paper title (planned): **T10.5 — Galois Structure of the Kummer Clock.**

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8. Coda

T4 worked the $m = 2$ rung of the gauge corridor: the multiquadratic radical clock K_N generated by the Helmert unit normalizers, with prime-rung growth. T10 lifts the perspective to the full corridor: every integer rung m has its own Kummer clock $K_N^{(m)}$, its own closure map π_m, its own conductor selection rule, its own polar partner $-m$, its own Kummer cover deck-group character ρ_m, and its own classical physical literatures. The corridor is a uniform structure parameterized by $m \in \mathbb{Z} \cup \{\pm\infty\}$.

The one-sentence apex of T10:

> **The gauge corridor of T0 §1.3 is a Kummer clock indexed by $m \in \mathbb{Z}$, with the Kummer clock field $K_N^{(m)} = \mathbb{Q}(\zeta_m, \sqrt[m]{p} : p \leq N)$ providing the natural arithmetic ambient at each rung; T4's multiquadratic radical clock is the $m = 2$ specialization, T6 the $m = 2$ worked closure, T3 the $m = 3$ worked selection rule, T5 the $|m| = 1$ polar pair, and the corridor extends to all integer m with the Kummer cover deck-group character $\rho_m(\zeta) = \zeta^{-m}$ controlling the integer-m Galois content.**

And the closing slogan:

> **The radical clock of T4 is the $m = 2$ rung. The full Kummer clock is the gauge corridor read through its arithmetic ambient.**

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